HotelRevAI – AI Driven Revenue Analysis for Hotels

Abstract

The Hotel Revenue Analysis project focuses on analyzing hotel booking data to evaluate business performance and identify key revenue drivers. The primary objective of this project is to extract meaningful insights from hotel reservation datasets and support strategic decision-making using data analytics techniques.

The dataset includes information such as booking dates, hotel type, customer segment, average daily rate (ADR), length of stay, cancellations, and other operational attributes. The data is first cleaned and preprocessed by removing duplicate records, handling missing values, eliminating invalid entries such as negative ADR values, and transforming date fields into structured formats for analysis.

Key performance indicators (KPIs) such as Total Revenue, Average Daily Rate (ADR), Revenue per Available Room (RevPAR), Occupancy Rate, and Cancellation Rate are calculated to measure hotel performance. Various analytical techniques are applied to examine trends across different months, years, hotel types, and market segments.

The project utilizes data visualization tools such as Power BI to create interactive dashboards that present revenue trends, customer behavior patterns, and segment-wise performance. These visualizations enable stakeholders to easily interpret data and make informed business decisions.

The findings of this project help identify peak revenue periods, high-performing customer segments, and factors affecting cancellations. Ultimately, the system assists hotel management in optimizing pricing strategies, improving occupancy rates, and enhancing overall profitability through data-driven decision-making.

1. Introduction

This project focuses on analyzing hotel revenue using a real-world booking dataset spanning **three years (2018–2020)**. Although the dataset duration is limited, it contains **detailed transactional, pricing, and customer information**, which makes it highly suitable for revenue analysis, feature engineering, and data cleaning practice.

The objective of this documentation is to explain:

- The **key dataset columns** critical for revenue analysis
- The **newly created columns** (derived features)
- The **logic and arithmetic operations** used to create these columns

2. Dataset Significance

The dataset represents real hotel booking behavior and includes:

- Pricing information (ADR)
- Stay duration (weekday and weekend nights)
- Cancellation behavior
- Market and distribution channels
- Customer segmentation

Such attributes allow the computation of industry-standard hotel KPIs like **Total Revenue**, **Occupancy-based Revenue**, and **Segment-wise Revenue Trends**.

3. Key Columns for Revenue Analysis

The following columns are the **most important for hotel revenue analysis**:

3.1 Pricing Column

- **adr** (Average Daily Rate):
  Represents the average price paid per room per night. It is the core variable used in revenue calculations.

3.2 Stay Duration Columns

- **stays_in_weekend_nights** – Number of weekend nights stayed
- **stays_in_week_nights** – Number of weekday nights stayed

These columns define how long a guest stays, which directly impacts total revenue.

3.3 Booking Status Column

- **is_canceled** – Indicates whether a booking was cancelled (1) or completed (0)

This column is crucial to ensure that revenue calculations consider **only realized bookings**.

3.4 Time-Related Columns

- **arrival_date_year**
- **arrival_date_month**
- **arrival_date_day_of_month**

These columns enable trend analysis, seasonality detection, and time-series aggregation.

3.5 Segmentation Columns

- **hotel** – Type of hotel (City Hotel / Resort Hotel)
- **market_segment** – Booking source category
- **distribution_channel** – Channel used for booking
- **customer_type** – Type of guest

These columns help identify which segments contribute the most to revenue.

4. Unnecessary / Less Relevant Columns for Revenue Analysis

Not all columns in the dataset contribute directly to hotel revenue analysis. Some columns are identifiers, personal information, or high-cardinality fields that do not provide meaningful insights into revenue generation. Including such columns can introduce noise and reduce analytical efficiency.

4.1 Characteristics of Unnecessary Columns

- Do not influence pricing, stay duration, or booking status
- Contain personal or identification information
- Have no direct or indirect relationship with revenue metrics

4.2 Examples of Unnecessary or Low-Impact Columns

- **Arrival_date_year, month, day**
- **Free-text remarks or comments** – Unstructured and inconsistent
- **Agent or company identifiers** – Low relevance unless agent-level analysis is required

4.3 Action Taken During Data Cleaning

- Such columns are **dropped or ignored** during analysis
- Retained only if required for reference or traceability

4.4 Significance of Removing These Columns

- Improves clarity of revenue analysis
- Reduces data noise
- Enhances model performance
- Increases computational efficiency

5. Newly Created Columns (Feature Engineering)

To enhance revenue analysis, several new columns are derived using arithmetic and logical operations.

5.1 Total Stay Nights

New Column: total_stay_nights

Creation Logic:

total_stay_nights = stays_in_weekend_nights + stays_in_week_nights

Significance:
Represents the total number of nights booked per reservation. It is required for computing booking revenue.

5.2 Realized Revenue

New Column: realized_revenue

Creation Logic:

realized_revenue = booking_revenue × is_effective_booking

Significance:
Represents the actual revenue earned by the hotel after excluding cancelled bookings.

5.3 Unified Arrival Date

New Column: arrival_date

Creation Logic:
Combine arrival_date_year, arrival_date_month, and arrival_date_day_of_month into a single date column.

Significance:

Enables chronological analysis, monthly revenue trends, and time-based visualization.

6. Project Execution Steps

Data Collection

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Data Cleaning

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Data Transformation

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KPI Calculation

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Analysis

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Dashboard Creation

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Business Insights

1. Load the dataset and inspect structure
2. Handle missing values and incorrect data types
3. Create derived revenue-related columns
4. Filter cancelled bookings for realized revenue
5. Aggregate revenue by year, month, hotel type, and segment
6. Perform exploratory data analysis (EDA)
7. Visualize revenue trends and patterns
8. Interpret insights and summarize findings

7. Conclusion

Despite covering only three years, the dataset provides sufficient depth for meaningful hotel revenue analysis. By identifying key columns and creating revenue-driven features through arithmetic operations, the project demonstrates strong analytical methodology, data preprocessing skills, and business relevance.